Prodigy Anesthesia Quick Review

Alzheimers/Dementia

What is the pathologic process that results in the neuronal degradation associated with Alzheimer's disease?	Deposition of amyloid beta peptides produces neuritic plaques and neurofibrillatory tangles that result in disruption of neurotransmitter function and the death of neurons. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 620.*
What test is diagnostic of Alzheimer's disease?	Although computed tomography may demonstrate ventricular dilation and cortical atrophy and positive emission tomography may exhibit areas of decreased cerebral blood flow, only postmortem examination of the brain tissue is definitively diagnostic of Alzheimer's. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 245.*
What is the incidence of Alzhemier's and what population does it affect the most?	Alzheimer's is the most common form of neurodegenerative disease and is responsible for 40-80% of all cases of dementia. It affects approximately 20% of of patients over 80 years of age. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 619.*
What is the standard medical treatment for patients with Alzheimer's disease?	Cholinesterase inhibitors such as rivastigmine, donepazil, and galantamine are the standard medical treatments for Alzheimer's disease. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 245.*
What are the side effects of the most common drugs used to treat Alzheimer's?	Acetylcholinesterase inhibitors are the most common treatment and may result in nausea, vomiting, bradycardia, syncope, and fatigue. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 620.*
How is the function of choline acetyltransferase affected in patients with Alzheimer's disease?	Patients with Alzheimer's experience a significant decrease in the activity of choline acetyltransferase. *Mccance KL, Huether SE, Brashers VL, & Rote NS. Pathophysiology: The Biologic Basis of Disease. 6th ed. Philadelphia, PA: Mosby, Inc.; 2010: 554.*
What are the anesthetic management implications for the patient with Alzheimer's disease?	Preoperative sedation is usually avoided because it may aggravate dementia and precipitate postoperative confusion. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 620.*

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Alzheimers/Dementia

How would you expect a patient with Alzheimer's to respond to succinylcholine?	Patients taking acetylcholinesterase inhibitors may have a prolonged duration of action with succinylcholine. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 621.*
How would you expect a patient with Alzheimer's disease to respond to muscle relaxants?	They will exhibit a resistance to nondepolarizing muscle relaxants due to the use of acetylcholinesterase inhibiting drugs as part of their medical management. They will also exhibit a longer duration of action with succinylcholine. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 245.*
If you need to administer an anticholinergic to a patient with Alzheimer's disease, which one is preferred and why?	Glycopyrrolate is preferred because it doesn't cross the blood-brain barrier. Atropine and scopolamine may result in increased confusion. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 621.*

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